

Project 2: Document Model

Goal: Solve a problem that requires creating a fully established secondary data set, $\mathbb{D}'$, using the document model.

Process:

1. Select a general problem from list below.
 2. Refine that general problem into a specific problem.
 3. Develop a specific solution (through iteration).
 - Create a Document Database (MongoDB)
 - Creating an Analysis and Visualization Pipeline (.ipynb)
 - Create Metadata to fully establish the data
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General Problems

1. Detecting credit card fraud
 2. Predicting loan default risk
 3. Forecasting stock prices
 4. Forecasting global climate change
 5. Detecting AI generated images/text/etc.
 6. Projecting athletic performance
 7. Predicting sports game outcomes
 8. Clinical drug trials
 9. Predicting hospital readmission risk
 10. Allocating emergency response resources
 11. Forecasting energy demand
 12. Predicting wildfire risk
 13. Predicting air quality
 14. Recommending content (e.g. Netflix)
 15. Detecting online bots
 16. Predicting election results
 17. Other (after office hours discussion)
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Rubric

Requirement	Yes	No	ψ
General			
Materials are submitted on time	☐	☐	1
Materials are submitted as a well organized GitHub repository	☐	☐	4
README a markdown file that makes all materials easy to find and access. In root folder of project. (Header structure indicated throughout this rubric.)	☐	☐	5
Coding Standards			
Project is written in Python, Markdown, and mongosh	☐	☐	1
All code runs without major errors	☐	☐	1
Code is commented inline to explain every class/function	☐	☐	1
Python includes proper error handling	☐	☐	1
Python includes logging to log files	☐	☐	1
Project Details - L1 Header			
Title - "DS 4320 Project 2:" followed by your project title, this is the L1 line of the README	☐	☐	1
Executive Summary - Short paragraph explaining the contents of the repository in executive form	☐	☐	3
Name - your name	☐	☐	1
NetID - your NetID	☐	☐	1
DOI - create a DOI for your project	☐	☐	1
Press Release - link to press release	☐	☐	1
Pipeline - link to pipeline file	☐	☐	1
License State name of license here and link to the file in the top level of the repository (use normal GitHub conventions)	☐	☐	1
Problem Definition - L2 Header			
State the initial general problem and refined specific problem statement	☐	☐	3
One paragraph explaining the motivation for the project	☐	☐	3
One paragraph explaining the rationale for the refinement from general to specific	☐	☐	3
Headline of Press Release and link to separate markdown file containing the press release	☐	☐	1
Domain Exposition - L2 Header			
Terminology - Table summarizing jargon, KPIs, etc.	☐	☐	2
Paragraph explaining the domain the project lives in.	☐	☐	2
Background reading Separate OneDrive folder containing copies of articles, blog posts, etc. about the project topic (1 point per, up to 5 points, can do more).	☐	☐	1
These should help the reader understand the domain of the project.	☐	☐	1

	□	□	1
	□	□	1
	□	□	1
Table - showing a summary of the readings, one row per item, includes title, brief description, and link to file in folder	□	□	1
Data Creation - L2 Header			
Paragraph (or two) explaining the raw data acquisition process (provenance).	□	□	2
Code Table showing the code used to create the data, one row per file, with a brief description and link to source code in repo	□	□	3
Rationale for critical decisions, especially judgement calls, and places that can introduce/mitigate uncertainty	□	□	3
Bias Identification description of how bias could be/was introduced in the data collection process	□	□	1
Bias Mitigation description of how biases can be handled/quantified/accounted for in analysis	□	□	1
Metadata - L2 Header			
Soft-Schema Guidelines for document structure	□	□	3
Data Summary Summary of Database contents - tabular form is permissible	□	□	1
Data Dictionary Table with one row per feature in the data set containing: name, data type, description, example	□	□	3
Data Dictionary quantification of uncertainty for numerical features	□	□	3
Press Release - separate markdown file			
Headline - L1 header	□	□	2
Hook - L2 header - Short, holds attention, explains the motivation	□	□	2
Problem Statement - L2 header - Describe the current state of affairs in more detail and explain your specific problem	□	□	2
Solution Description - L2 header - Explain your solution but keep it high level, the target is the end-user not a technical person	□	□	2
Chart - L2 Header - A chart that visualizes your data and supports your solution	□	□	2
Data - Stored In Mongo Atlas			
Scale Dataset contains more than 10 documents	□	□	1
Scale Dataset contains more than 100 documents	□	□	2
Scale Dataset contains more than 1000 documents	□	□	3

In comment in Canvas a username and password for your grader to use for accessing your DB through mongosh (do not put on GitHub).	□	□	4
Problem Solution Pipeline - Separate Files			
File 1: create your pipeline in a jupyter notebook	□	□	1
File 2: also save the notebook as a markdown file	□	□	1
Data preparation query MongoDB to put data into a dataframe	□	□	2
Solution analysis Implement a model	□	□	1
Analysis rationale explain the decisions in your analysis process	□	□	1
Analysis complexity Utilize ML/AI taught in DS 3021/4021	□	□	1
Visualize results Create a visualization to show your results	□	□	1
Visualization rationale explain the decisions in your visualization process	□	□	1
Visualization is publication quality	□	□	1
Pipeline solves the problem	□	□	5
Total			100